GEOPHYSICAL RESEARCH LETTERS, VOL. 11, NO. 12, PAGES 1231-1234, DECEMBER 1984

CLINKER DEPOSITS, POWDER RIVER BASIN, WYOMING AND MONTANA: A NEW SOURCE OF HIGH-FIDELITY PALEOMAGNETIC DATA FOR THE QUATERNARY

Alison H. Jones[1], John W. Geissman[2]

Department of Geology, Colorado School of Mines

Donald A. Coates

U.S. Geological Survey, Denver

Abstract. Clinker, rock baked and fused by in situ burning of underlying coal, posesses a thermoremanent and/or thermochemical magnetization that resides in magnetite, hematite, and goethite. Data from individual sites imply that clinker may provide a high resolution recording of the geomagnetic field (e.g., within-site virtual geomagnetic pole angular standard deviations are usually less than 10°), but this is dependent on its thermal history and the time over which magnetic phases formed. Data from 17 sites in the Powder River Basin (13 normal polarity, 4 reverse) yield a mean of Decl. = 348°, Incl. = 62°, α_{95} = 5°, k = 42 and a corresponding pole position of 81° lat., 158° long. with a virtual geomagnetic pole angular standard deviation of 17.3°. The mechanism of clinker formation, the wide geographic distribution, and the range of available fission-track ages suggest that clinker may contain a nearly complete geomagnetic field record for much of the Quaternary.

Introduction

Geomagnetic field variations that have occurred during the past five Myr have been investigated using marine and terrestrial sedimentary rocks, lava flows, and baked clays, as summarized by Barton and Merrill [1983], Creer et al. [1983], and Merrill and McElhinny [1983]. Problems and/or limitations exist in the paleomagnetic analysis of all of these units. For volcanic flows and baked clays, time coverage is discontinuous, so that time between magnetization events is difficult to estimate. For sedimentary rocks, the natural remanent magnetization (NRM) may not always faithfully reproduce the directions of the ambient magnetic field, as discussed by Verosub and Banerjee [1977] and Barendregt [1981].

In an attempt to establish a new source of high-fidelity data for the Quaternary, we have investigated clinker--rock that has been baked and fused during intermittent near-surface burning of coal beds [Rogers, 1917; Coates, 1980]. Clinker of Quaternary age is abundant in several Tertiary basins in the Rocky Mountains. Although unorthodox for paleomagnetic studies, clinker, based on our results, appears to be an acceptable geomagnetic field recorder. Because detrital zircons in the baked sediments were annealed during heating, Coates and Naeser [1984] showed that fission-track ages can be used to give the age of baking, and thus the approximate age of magnetizations, although analytical uncertainties are relatively high and direct correlation with paleomagnetic data may be tenuous. In the eastern Powder Basin, clinker has formed where Tertiary coal beds that dip gently basinward have been exposed by erosion. The extensive clinker deposits represent a number of discrete burn events. Up dip, clinker is progressively older, with fission-track ages ranging from less than 0.08 to 0.77 ± 0.39 my [Coates and Naeser, 1984]. In the northern Powder River Basin, ages are as old as 1.4 my [Jones et al., 1982]. If individual burn events occurred intermittently and were of a few tens to a few hundreds of years duration, individual clinker deposits may contain an almost continuous Quaternary field record.

Few reports describe remanence and/or rock magnetic properties of clinker [Krs, 1968; Purucker, written comm., 1980; Watson, 1978]. Our study in the Powder River Basin is intended to evaluate whether clinker, albeit complicated from a rock magnetic and remanence standpoint, is capable of providing high-quality geomagnetic field information, including, as a part of future studies, paleointensity data for fused material. The paleomagnetic data reported are from five areas chosen on the basis of fission-track ages or particularly good exposure.

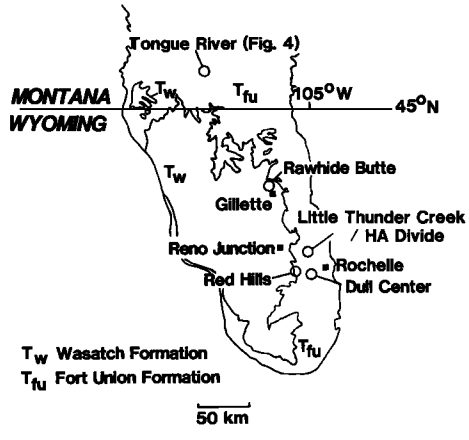


Fig. 1. Generalized map of the Powder River Basin. Circles show sampling locations. The Paleocene Fort Union and the Eocene Wasatch Formations contain numerous clinker deposits.

[1] Geological Consultant, 1645 S. Pennsylvania, Denver, Colorado 80210

[2] Department of Geology, University of New Mexico, Albuquerque, New Mexico 87131

Copyright 1984 by the American Geophysical Union.

Paper number 4L6324.
0094-8276/84/004L-6324$03.00

Sampling and Methodology

A total of 150 samples were collected at 17 sites in five areas in the northern and eastern Powder River Basin (Figure 1). Sampling was confined to areas where clinker was baked to a brick-like appearance and was least deformed (slumped during collapse). To date, we have not investigated fused material. Dip of the bedding is two degrees or less at all sites except two.

The fractured nature of the clinker required collection of samples (generally 7 or more per site) as oriented blocks from which specimens were extracted for laboratory study. Data were recorded with a spinner magnetometer modified with an optical shaft encoder and interfaced with a microprocessor. Both alternating field and thermal demagnetization were carried out using commercially available units. Directions of all magnetization components were determined using vector subtraction. Rock magnetism investigations consisted of isothermal remanent magnetization (IRM) acquisition, backfield DC demagnetization, and Curie temperature (saturation magnetization, J_s vs. T) analysis.

Results

Median destructive fields for clinker generally exceeded 50 milliTesla (mT), and more than 80 percent of the samples contained a single component NRM, which was best resolved with thermal demagnetization (Figure 2a, b). Blocking temperature spectra vary considerably, probably as a result of a complicated and variable magnetic mineralogy. Samples displaying multicomponent behavior (Figure 2c) were found at many sites, and components characteristic of a site are not always removed over a constant range of temperature. Thermal demagnetization was done progressively; heating increments were dictated by the observed response.

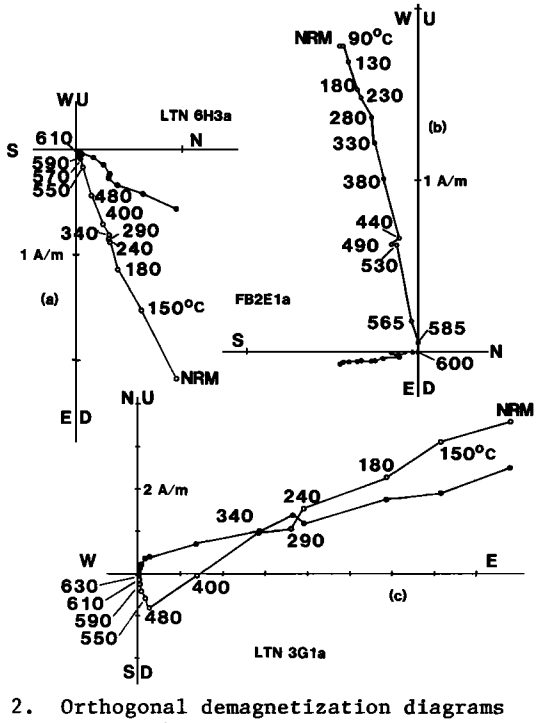


Fig. 2. Orthogonal demagnetization diagrams (Zijderveld, 1967) showing the removal of magnetization components in clinker during thermal treatment. Closed (open) symbols refer to horizontal (vertical) projections. Demagnetizations of (a) single component, normal polarity clinker; (b) single component, reverse polarity clinker; and (c) multicomponent, normal polarity clinker.

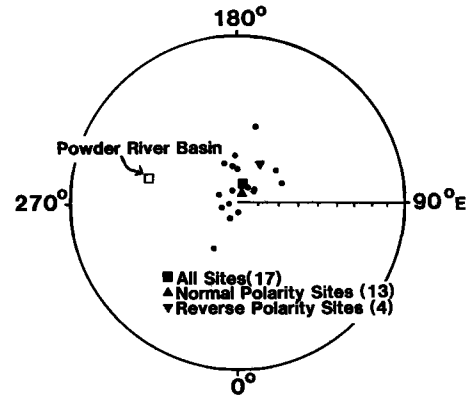


Fig. 3. Equal area stereographic projection of the 17 site mean directions and projected cones of 95% confidence. Excluding some corrections for local slumping effects (i.e., site DC2, see text), the directions are plotted with respect to geographic coordinates. Solid (open) symbols refer to lower (upper) hemisphere projections.

Site mean directions, plotted in Figure 3, do not cluster about the present-day field. A number of discrete burn events during the Quaternary appear to have been recorded by individual sites. At four sites up to three samples were omitted in the statistical analysis because of extremely anomalous directions. Results from four localities are briefly described.

Samples were collected at five sites (BDS1,2,3 and FB1,2) along and to the northwest of a promontory above the Tongue River near Ashland, Montana (Figure 4). Sites BDS1,2,3 and FB1 are the four reverse polarity sites in Figure 3. The samples from site FB2, in the clinker judged youngest as the site is farthest downdip, are of normal polarity. The reverse polarity sites must record burning prior to the Matuyama-Brunhes transition at about 730 ky. Jones et al. [1982] report a fission track age of 1.4 ± 0.36 my for clinker at site BDS2. On this basis we suggest that burning at the four reverse polarity sites took place prior to the Jaramillo Event; site FB2 possibly recorded the Jaramillo Event.

Three sites (RHI1-3) were collected in the Red Hills area (Figure 1). Sites are separated by at least 3 km. All samples contained only normal polarity components and the three site means are indistinguishable at the 95% confidence level.

Two sites (LTN1,2), approximately 100 m apart, were sampled along HA Divide north of Little Thunder Creek (Figure 1). Both sites show only normal polarity magnetizations in agreement with a fission-track age of 0.6 ± 0.29 my at site LTN1, but the site mean directions are statistically distinct.

Two sites (LTN3,4) were sampled on a narrow promontory projecting north of HA Divide, about 4

19448007, 1984, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL011i012p01231 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

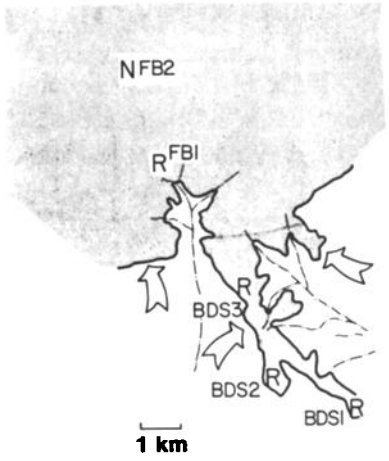


Fig. 4. Sketch map of the Tongue River sampling areas (Figure 1), showing the form of the clinker promontory sampled and the location of reverse (R) and normal (N) polarity sites. Arrows show the general direction of burning.

km northwest of sites LTN1,2. This area was examined because one of us (D.A.C.) found reverse polarity NRM using a portable fluxgate magnetometer. At LTN 3, slumping during burning complicated the structure, and a low unblocking temperature secondary component, often of negative inclination, is present in many of the samples (Figure 2b). The higher blocking temperature component (removed up to and past maximum magnetite blocking temperatures) is well-grouped and of normal polarity for both sites.

Rock Magnetism

The essential magnetic phases in clinker are low-Ti titanomagnetite (Curie temperatures range from 400 to 520°C and IRM curves show a distinct plateau at 0.1 to 0.3 T), hematite (commonly 10% of the NRM remains unblocked well above 580°C) and goethite (in many cases up to 40% of the NRM is removed by 250°C). The cubic and rhombohedral phases appear to be stable from a crystal-chemical standpoint in that low field susceptibility remains constant in thermal demagnetization up to 680°C and J_s-T runs are reversible. We suggest that titanomagnetite forms in a manner analogous to that described by Stephenson [1967] and Dunlop [1972] in which hematite- and ferromagnesian silicate-bearing sandstones are thermally demagnetized in air. We thus infer that the essential magnetization in clinker is a thermoremanent magnetization (TRM) residing in a cubic phase. Hematite and goethite may contain thermochemical remanent magnetizations (TCRM) acquired during cooling of clinker and possibly reflecting groundwater movement. If these magnetizations do not significantly complicate the NRM, high resolution secular variation data may be obtained.

Discussion

Remanence data and rock magnetic tests indicate that much of the (characteristic) magnetization in clinker was acquired during cooling. The data in Figure 3 are consistent with a time-averaged Quaternary field direction, but the mean of the 17 site means is of a more westerly declination.

We now evaluate the dispersion of clinker data in terms of secular variations (SV) of the field. TRM acquisition in clinker may be similar to that in baked clays [e.g., Creer et al., 1983], depending largely upon the length of thermal heating and thus magnetization acquisition at a particular location. With a sufficient data base, high resolution estimates of field variations over time may be obtainable. Multiple magnetic phases, some of which may contain chemical overprints, and variable blocking temperature ranges may, however, actually cause the SV record in clinker to be smeared. Within-site paleomagnetic data suggest that these effects are not significant enough to preclude the recording of some high resolution SV information. VGP angular standard deviation (S_w) values for each of 13 normal polarity sites range from 5.1° to 11.5°. These values are lower than a 15.6° value predicted by Cox's [1970] field secular variation model D for 45° latitude or the 16.6° value given by model F of McFadden and McElhinny [1984], although they are greater than those typical of young, rapidly cooled lavas. Individual clinker sites probably cooled more rapidly than those in intrusions, however, and we expect that, in most cases, clinker acts more like an "instantaneous" field recorder than intrusive rocks. We note that each reverse polarity site yields S_w values ranging from 12.0° to 16.3°. Until more reverse polarity data are available, the greater VGP dispersion for the reverse polarity sites cannot be evaluated in terms of possible true "asymmetry" in field behavior [Merrill and McElhinny, 1977; Merrill et al., 1979]. Straightforward response in demagnetization precludes incomplete magnetic cleaning as the cause of the increased scatter. The VGP angular standard deviation for the mean of the 17 site means is 17.3°; for the 13 normal polarity sites it is 15.8°; and for the 4 reverse polarity sites it is 17.2°.

Potential contributors to the within-site dispersion (S_w) of clinker data, which would be different from those for lavas, are (1) gradual to abrupt overburden collapse during coal burning and/or cooling, (2) rates of cooling which are slow in relation to SV, and (3) multicomponent NRM residing in several magnetic phases. Laboratory unblocking temperatures are generally distributed, and clinker collapse at moderate temperatures may cause dispersion of the already blocked partial TRM, as suggested by Purucker (written comm., 1980). At site DC2 in the Dull Center area (Figure 1), we have found this to be true. Directions of high (>540°C) blocking temperature magnetizations from a 4 m wavelength collapse-related syncline yield a $K_{corrected}/k_{in\ situ}$ ratio of 5.5 for 8 samples. An F-ratio test establishes that at the 99% probability level blocking occurred prior to folding. If the NRM in clinker is simply the sum of PTRM's, dispersion of directions during progressive thermal demagnetization, especially if it can be related to the orientation of collapse structures, is compelling evidence for collapse as the cause of the dispersion. Systematic directional changes with progressive demagnetization at an undeformed site may reflect slow cooling. In the case of multicomponent NRM,

19448007, 1984, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL011i012p01231 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

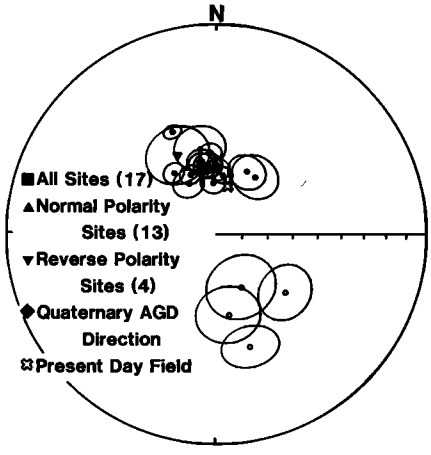


Fig. 5. North polar projection of VGP site and VGP mean (from Figure 3) data.

we have found that lower blocking components (e.g., as shown in Figure 2c) are more dispersed than high blocking temperature components. These components are most likely partial TRM's or TCRM's acquired during or long after collapse; they may have anomalous directions (e.g., Figure 2c). Dispersion caused by collapse would be dependent both on the ambient temperature and the blocking temperature spectra of the magnetization(s). These factors probably vary widely from site to site.

The clinker data, shown as VGP positions in Figure 5, do not support a dipole offset model for the time-averaged geomagnetic field [Wilson, 1971; Merrill and McElhinny, 1977]. We note, however, the remark by Barton and Merrill [1983] that such models must be tested with global data.

The formation of clinker in the Powder River Basin has been an almost continuous process throughout much of the Quaternary. Ongoing work involves detailed sampling for paleomagnetic and paleointensity data where burning is active today, at sites which, on the basis of fission track ages and/or physiography, apparently burned during the Matuyama epoch, and at sites of intermediate ages.

Acknowledgements. Support by two G.S.A. grants and one Sigma Xi grant to A.H.J and a D.O.E. contract to J.W.G is appreciated. We thank Carter Mining Company, and in particular F.W. Metzger, for summer logistical support. J_s-T measurements were made at the U.S.G.S. laboratory, Denver. J.G. Rosenbaum and W.P. Hasbrouck provided pre-submittal reveiws.

References

Barendregt, R.W., Dating methods of Pleistocene deposits and their problems: VI. Paleomagnetism, Geoscience Canada, 8, 56-64, 1981.

Barton, C.E., and R.T. Merrill, Archaeo- and paleosecular variation, and long-term asymmetries of the geomagnetic field, Rev. Geophys. Space Phys., 21, 603-614, 1983.

Coates, D.A., Formation of clinker by natural burning of coal beds in the Powder River Basin, Wyoming, in Wyoming Geological Association Guidebook, Powder River Basin, edited by J. Strickland, 45-48, 1980.

Coates, D.A., and C.W. Naeser, Map showing fission-track ages of clinker in the Rochelle Hills, southern Campbell County, Wyoming, U.S. Geol. Surv. Miscl. Inv. Map I-1462, 1984.

Cox, A.V., Latitude dependence of the angular dispersion of the geomagnetic field. Geophys. J. Roy. Astr. Soc., 20, 253-269, 1970.

Creer, K.M., P. Tucholka, and C.E. Barton (Eds.). Geomagnetism of Baked Clays and Recent Sediments, 324 pp., Elsevier, Amsterdam, 1983.

Dunlop, D.J., Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis, Geophys. J. Roy. Astr. Soc., 27, 37-55, 1972.

Jones, A.H., J.W. Geissman, D.A. Coates, C.W. Naeser, and E.L. Heffern, Paleomagnetism of Powder River Basin clinker, Wyoming and Montana: documentation of Bruhnes and Matuyama burning of Tertiary coals (abstract), EOS, Trans. Amer. Geophys. Union, 63, 304, 1982.

Krs, M., Magnetisierung von thermal umgewandelten Gesteinen im Braunkohlenbecken von Chomutov-Most-Templice, Vestnik Ustredniho Ustavu Geo., 43, 359-362, 1968.

McFadden, P.L., and McElhinny, M.W., A physical model for paleosecular variation, Geophys. J. Roy. Astr. Soc., in press, 1984.

Merrill, R.T., and McElhinny, M.W., Anomalies in the time-averaged paleomagnetic field and their implications for the lower mantle, Rev. Geophys. Space Phys., 15, 309-323, 1977.

Merrill, R.T., and M.W. McElhinny, The Earth's Magnetic Field, 401 pp., Academic Press, London, 1983.

Merrill, R.T., M.W. McElhinny, and D.J. Stevenson, Evidence for long-term asymmetries in the earth's magnetic field and possible implications for dynamo theory, Phys. Earth Planet. Int., 20, 75-82, 1979.

Rogers, G.S., Baked shale and slag formed by the burning of coal beds, U.S. Geol. Surv. Prof. Paper 108-A, 1-10, 1917.

Stephenson, A., The effect of heat treatment on the magnetic properties of the Old Red Sandstone, Geophys. J. Roy. Astr. Soc., 13, 425-440, 1967.

Verosub, K.L. and S.K. Banerjee, Geomagnetic excursions and their paleomagnetic record, Rev. Geophys. Space Phys., 15, 145-155, 1977.

Watson, D.E., Magnetic properties of clinkers (abstract), Geophysics, 44, 376-377, 1978.

Wilson, R.L., Dipole offset -- the time averaged paleomagnetic field over the past 25 million years, Geophys. J. Roy. Astr. Soc., 22, 491-504, 1971.

Zijderveld, J.D.A., A.C. Demagnetization of rocks: Analysis of results, in Methods in Paleomagnetism, edited by D.W. Collinson, K.M. Creer, and S.K. Runcorn, pp. 254-286, Elsevier, 1967.

D.A. Coates, U.S. Geological Survey, Mail Stop 972, Denver Federal Center, Denver, CO 80225.
J.W. Geissman, Department of Geology, University of New Mexico, Albuquerque, NM 87131.
A.H. Jones, Geological Consultant, 1645 S. Pennsylvania, Denver, CO 80210

(Received August 17, 1984;
revised October 9, 1984;
accepted October 15, 1984.)

19448007, 1984, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL011i012p01231 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License